Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
aabasangova@aabasangova:~$
aabasangova@aabasangova:~$ ls /etc/ > file.txt
aabasangova@aabasangova:~$ ls >> file.txt
aabasangova@aabasangova:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
aabasangova@aabasangova:~$ grep .conf file.txt > conf.txt
aabasangova@aabasangova:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/aabasangova/.npm/_cacache/index-v5/09/c0
/home/aabasangova/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc
29def30220242c53
/home/aabasangova/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad5
5fb2000f59e2edcc
/home/aabasangova/.npm/_cacache/index-v5/cd
/home/aabasangova/.npm/_cacache/index-v5/c9
/home/aabasangova/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea31
0cda0b203217f693
/home/aabasangova/.npm/_cacache/index-v5/c0
/home/aabasangova/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d135
0482a1d990504757
/home/aabasangova/.npm/_cacache/index-v5/af/cf
/home/aabasangova/.npm/_cacache/index-v5/c3
/home/aabasangova/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f25062
14f94db8861160d3
/home/aabasangova/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c818
8a44d554f3f3ba2e
/home/aabasangova/.npm/_cacache/index-v5/ca
/home/aabasangova/snap/hugo/common
/home/aabasangova/snap/hugo/current
/home/aabasangova/conf.txt
aabasangova@aabasangova:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
aabasangova@aabasangova:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
aabasangova@aabasangova:~$
aabasangova@aabasangova:~$ find ~ -name "log*" > logfile &
[1] 6629
aabasangova@aabasangova:~$
[1]+  Done                    find ~ -name "log*" > logfile
aabasangova@aabasangova:~$ rm logfile
aabasangova@aabasangova:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
aabasangova@aabasangova:~$ gedit &
[1] 6672
aabasangova@aabasangova:~$ ps | grep gedit
   6672 pts/0    00:00:00 gedit
aabasangova@aabasangova:~$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec
]
aabasangova@aabasangova:~$ kill 6672
[1]+  Terminated              gedit
aabasangova@aabasangova:~$
aabasangova@aabasangova:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual page documents the GNU version of df.  df displays the amount of space
       available on the file system containing each file name argument.  If no  file  name
       is  given,  the  space  available  on  all currently mounted file systems is shown.
       Space  is  shown  in  1K  blocks  by  default,  unless  the  environment   variable
       POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file
       system,  df  shows  the space available on that file system rather  than on the file
       system containing the device node.  This version of df cannot show the space avail-
       able on unmounted file systems, because on most kinds of systems doing so  requires
       non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file sys-
       tems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
              of 1,048,576 bytes; see SIZE format below
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print apparent sizes rather than device usage; although the apparent size is
              usually smaller, it may be larger due to holes in ('sparse') files, internal
              fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes by SIZE before printing them; e.g., '-BM' prints sizes in units
              of 1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
aabasangova@aabasangova:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 90469716  64073772  59% /
devtmpfs         4011288        0   4011288   0% /dev
tmpfs            4035660      100   4035560   1% /dev/shm
tmpfs            1614268     1968   1612300   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p3 155186176 90469716  64073772  59% /home
tmpfs            4035664     1644   4034020   1% /tmp
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807132      176    806956   1% /run/user/1048
/dev/sr0         2677920  2677920         0 100% /run/media/aabasangova/Fedora-WS-Live-43
aabasangova@aabasangova:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4        ./.npm/_cacache/index-v5/d4
4        ./.npm/_cacache/index-v5/30/1a
4        ./.npm/_cacache/index-v5/30
424      ./.npm/_cacache/index-v5
115628   ./.npm/_cacache
115692   ./.npm
16       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
16       ./snap/hugo/25676/.config/go/telemetry
16       ./snap/hugo/25676/.config/go
16       ./snap/hugo/25676/.config
16       ./snap/hugo/25676
0        ./snap/hugo/common
20       ./snap/hugo
20       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1516296  .
aabasangova@aabasangova:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/aabasangova/.npm/_cacache/index-v5/a7
/home/aabasangova/.npm/_cacache/index-v5/a7/81
/home/aabasangova/.npm/_cacache/index-v5/d4
/home/aabasangova/.npm/_cacache/index-v5/d4/09
/home/aabasangova/.npm/_cacache/index-v5/30
/home/aabasangova/.npm/_cacache/index-v5/30/1a
/home/aabasangova/snap
/home/aabasangova/snap/hugo
/home/aabasangova/snap/hugo/25676
/home/aabasangova/snap/hugo/25676/.config
/home/aabasangova/snap/hugo/25676/.config/go
/home/aabasangova/snap/hugo/25676/.config/go/telemetry
/home/aabasangova/snap/hugo/25676/.config/go/telemetry/local
/home/aabasangova/snap/hugo/25676/.config/go/telemetry/upload
/home/aabasangova/snap/hugo/common
/home/aabasangova/monthly
/home/aabasangova/reports
/home/aabasangova/reports/monthly
/home/aabasangova/reports/monthly/monthly
/home/aabasangova/ski.plases
/home/aabasangova/ski.plases/equipment
/home/aabasangova/ski.plases/plans
/home/aabasangova/australia
/home/aabasangova/play
/home/aabasangova/play/games
/home/aabasangova/play/games/play
aabasangova@aabasangova:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.